

Statistical tests and assumptions

Project: Reproducible AI Research Study

1. Your mission

Your learning target

I can explain the mechanism and terminology of statistical tests and assumptions.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of statistical tests and assumptions

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Effect size and practical importance. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

My starting explanation

This week's end goal

Explain statistical tests and assumptions using the records and boundaries of this fictional investigation.

2. Notice the evidence

A statistical test compares observed evidence with a specified null model. A p-value is not the probability that the null hypothesis is true.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

List the given facts and one fact that is not supplied.

3. See how it works

Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all 2^n sign assignments, holding magnitudes fixed. Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.

Statistical tests and assumptions — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Differences 1, 1, 1 have observed mean 1. Enumerate 8 assignments.

Check or calculation: Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all 2^n sign assignments, holding magnitudes fixed. Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.

Conclusion and limit: Only all-plus and all-minus have absolute mean 1; others have $1/3$. Exact $p = 2/8 = .25$. At alpha .05 do not reject; this does not prove zero effect.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Differences1, 1, 1 have observed mean1. Enumerate8 assignments. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Break the task pairing and use unrelated A and B measurements.

Prediction: This paired sign-flip calculation is no longer justified by the stated design.

Reason: Its differences must come from the same task, with the declared independent sign-symmetry assumptions.

5. Words that unlock the idea

Word · Meaning

Word: Null model

Meaning: The probability model used for a test comparison

Word: P-value

Meaning: The null-model chance of an outcome at least as extreme under the defined statistic

Word: Exchangeability

Meaning: An assumption permitting the declared rearrangements

Use two terms to explain a step in this week's example.

6. Follow a worked example

Differences1, 1, 1 have observed mean1. Enumerate8 assignments.

Only all-plus and all-minus have absolute mean1; others have1/3. Exact $p=2/8=.25$. At alpha .05 do not reject; this does not prove zero effect.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary input, record the outcome and state the contract or statistical assumptions. The program uses fictional values and the Python standard library; it performs no external operations.

```
from itertools import product

def sign_flip_p(differences):

    """Two-sided exact test under independent symmetric sign
    exchangeability.

    Tiny n only; hold absolute magnitudes fixed and include equal extremes.

    A p-value is not a posterior probability that the null is true.

    """
```

```

if not differences or len(differences) > 16:
    raise ValueError('require 1-16 paired units')

threshold = abs(sum(differences))

count = 0

for signs in product((-1, 1), repeat=len(differences)):
    simulated = abs(sum((s * abs(d) for (s, d) in zip(signs,
differences))))

    if simulated >= threshold - 1e-12:
        count += 1

return count / 2 ** len(differences)

print(sign_flip_p([1, 1, 1]))

```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Statistical tests and assumptions

A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability.

The facts we will use

A times [5,8,4] seconds; B [4,6,3]. Differences B–A are [-1,-2,-1]. Use a two-sided statistic: absolute sum of differences. Assume the stated sign-symmetry null is appropriate.

Three independent paired task times

	A seconds	B seconds	B-A seconds
Task 1	5	4	-1
Task 2	8	6	-2
Task 3	4	3	-1

Read a cell using both its row and column labels.

Statistical tests and assumptions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Preserve paired differences

Differences are -1,-2,-1; their sum is -4 and mean is $-4/3$ seconds.



2. Enumerate null sign flips

Eight equally likely flips give sums -4,-2,0,2,4 with counts 1,2,2,2,1.



3. Count equally extreme sums

Only -4 and 4 have absolute sum at least 4: exact two-sided $p=2/8=0.25$.

CHANGE ONE THING

Break the task pairing and use unrelated A and B measurements.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Preserve paired differences

What happens and why: Differences are -1,-2,-1; their sum is -4 and mean is $-4/3$ seconds.

Step: 2. Enumerate null sign flips

What happens and why: Eight equally likely flips give sums -4,-2,0,2,4 with counts 1,2,2,2,1.

Step: 3. Count equally extreme sums

What happens and why: Only -4 and 4 have absolute sum at least 4: exact two-sided $p=2/8=0.25$.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A test compares an observed statistic with outcomes under a specified null model. Here the sign-flip assumptions make eight arrangements equally likely. At a predeclared 0.05 threshold, $p=0.25$ does not reject the null; it does not prove no difference.

What this example does not establish

Three pairs provide coarse evidence. The p-value is not the probability that the null is true; invalid independence or symmetry assumptions invalidate this test.

Read the labelled picture: Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p , and explain the decision at threshold 0.05.

Change one thing: Break the task pairing and use unrelated A and B measurements. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

The test requires its sign-exchangeability and independence assumptions; do not apply it blindly to clustered or asymmetric differences.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: The test requires its sign-exchangeability and independence assumptions; do not apply it blindly to clustered or asymmetric differences. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Statistical tests and assumptions”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Statistical tests and assumptions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p , and explain the decision at threshold 0.05.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Differences1, 1, 1 have observed mean1. Enumerate8 assignments. Show the intermediate steps and give the conclusion with its scope.

```
from itertools import product

def sign_flip_p(differences):
    """Two-sided exact test under independent symmetric sign
```

```
exchangeability.
```

```
Tiny n only; hold absolute magnitudes fixed and include equal extremes.
```

```
A p-value is not a posterior probability that the null is true.
```

```
"""
```

```
if not differences or len(differences) > 16:
```

```
    raise ValueError('require 1-16 paired units')
```

```
threshold = abs(sum(differences))
```

```
count = 0
```

```
for signs in product((-1, 1), repeat=len(differences)):
```

```
    simulated = abs(sum((s * abs(d) for (s, d) in zip(signs,
differences))))
```

```
    if simulated >= threshold - 1e-12:
```

```
        count += 1
```

```
return count / 2 ** len(differences)
```

```
print(sign_flip_p([1, 1, 1]))
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: The test requires its sign-exchangeability and independence assumptions; do not apply it blindly to clustered or asymmetric differences. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled statistical tests and assumptions evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of statistical tests and assumptions. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For the Reproducible AI Research Study, save paired differences, the null assumptions, all eight sign arrangements and the decision rule.

12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Statistical tests and assumptions — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Differences 1, 1, 1 have observed mean 1. Enumerate 8 assignments.

Check or calculation: Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all 2^n sign assignments, holding magnitudes fixed. Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.

Conclusion and limit: Only all-plus and all-minus have absolute mean 1; others have $1/3$. Exact $p = 2/8 = .25$. At alpha .05 do not reject; this does not prove zero effect.

The figure connects this chapter’s worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Apply a new case independently

Differences2, 2, 2 under the same null. Compute two-sided exact p and interpret.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Differences2, 2, 2 under the same null. Compute two-sided exact p and interpret.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a

changed case, not a copied answer.

15. Your lesson route

Start with this question: A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference’s sign to flip independently with equal probability.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of statistical tests and assumptions

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of statistical tests and assumptions

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only -4 and 4 have absolute sum at least 4: exact two-sided $p=2/8=0.25$.

1. Enumerate null sign flips
2. Count equally extreme sums
3. Preserve paired differences

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Count equally extreme sums Only -4 and 4 have absolute sum at least 4: exact two-sided $p=2/8=0.25$. A test compares an observed statistic with outcomes under a specified null model. Here the sign-flip assumptions make eight arrangements equally likely. At a predeclared 0.05 threshold, $p=0.25$ does not reject the null; it does not prove no difference.

Practice 2: explain the deciding evidence

Changed-case practice: Break the task pairing and use unrelated A and B measurements. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This paired sign-flip calculation is no longer justified by the stated design.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: This paired sign-flip calculation is no longer justified by the stated design. Its differences must come from the same task, with the declared independent sign-symmetry assumptions.

Repair a missing building block

Review Effect size and practical importance (ai-7/13). Goal: Explain the mechanism and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Review Sampling variability (ai-7/11). Goal: Explain the mechanism and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the mechanism and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.